SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	R. Nandhini	**Reg. No.:**	192524l355
Section / Batch:		**Date:**	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I ________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

CO2 - AT1

1> Closest pair - Brute force analysis :

1. problem Interpertation:

* Find the two points with the minimum Euction distance among the given points using the Brute force

Given points:

* A (2, 3), B (12, 30), C (40, 50), D(5, 1), E (2, 10), F(3, 4)

2. Justification of method :

⇒ The dataset contains only 6 points

⇒ Burte fore is simple and checks every possible pair

⇒ It guarantees the corect closet pair.

3. Distance Calculation:

Pair	Distance
AB	$\sqrt{(10^2+27^2)} = \sqrt{829} = 28.79$
AC	$\sqrt{(38^2+47^2)} = \sqrt{3653} = 60.44$
AD	$\sqrt{(3^2+(-2)^2} = \sqrt{13} = 3.61$
AE	$\sqrt{(10^2+7^2)} = \sqrt{149} = 12.21$

AF

$\sqrt{(1^2+1^2)} = \sqrt{2} = 1.41$

BC

$\sqrt{(-7)^2 + (-29)^3)} = \sqrt{890} = 29.83$

BD

$\sqrt{(-7) + (29)^2)} = \sqrt{890} = 29.83$

BE

$\sqrt{0^2 + (-20)^2)} = 20$

CD

$\sqrt{(-35)^2 + (-49)^2} = \sqrt{3626} = 60.22$

CE

$\sqrt{(-28)^2 + (-40)^2)} = \sqrt{2384} = 48.83$

CF

$\sqrt{(-37^2) + (46^2)} = \sqrt{3485} = 59.03$

DE

$\sqrt{(7^2+9^2)} = \sqrt{130} = 11.40$

DF

$\sqrt{(-2)^3+3^2)} = \sqrt{13} = 3.61$

EF

$\sqrt{(-9)^2 + (-6)^2} = \sqrt{117} = 10.82$

4) closest pair:

⇒ Points: A(2,3) and F(3,4)

⇒ Mininum Distance $= \sqrt{2} = 1.41$

5) Number of comparisons:

$$\frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15$$

Total comparisons $= 15$

6. Time complexity

→ Best case : $O(n^2)$

→ Average case : $O(n^2)$

→ Worst case : $O(n^2)$

7. Verification

* Among all calualed 1.411 is the smallest Hence (2, 3) and (3,4) are corectly identified as the closet pair

8. Real - world Application

→ GPS navigation

→ collision detection

→ Done path planning

→ Geographic information systems (GIS)